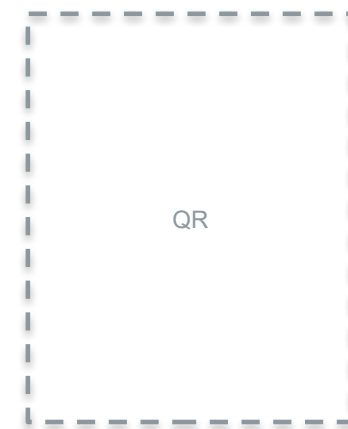


PLAS-Net: Lightweight Instance Segmentation of Beach Litter in Unmanned Aerial Vehicle Imagery

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Abstract

We present PLAS-Net, a lightweight model that returns an individual mask for every litter item in unmanned aerial vehicle imagery.

Sand, wrack and vegetation mimic weathered debris, so PLAS-Net adds dynamic feature fusion, context anchor attention and multi-scale sequence fusion to a compact segmentation baseline.

We compare it with eleven segmentation models, ablate each module in turn, and use its masks for scene-wide area summaries.

Motivation

- Bounding box: background sand counted as litter
- Semantic mask: touching items merged
- Instance mask: position, size and shape per item



Fig. 1. One tile drawn three ways: boxes, semantic map, instance masks.

Survey and Dataset

Site	Aow Luek Bay, Koh Tao, Thailand
Platform	DJI Mini 3, RGB
Altitude	5 m
Date	February 2024
Resolution	0.17 cm per pixel
Extent	59 × 91 m
Tiles	1,300 at 512 × 512
Split	905 fit, 395 evaluation
Annotation	2 annotators, 15 % cross-checked

Table 1. Survey acquisition and annotation.



Fig. 2. Processing chain from flight to annotated tiles.

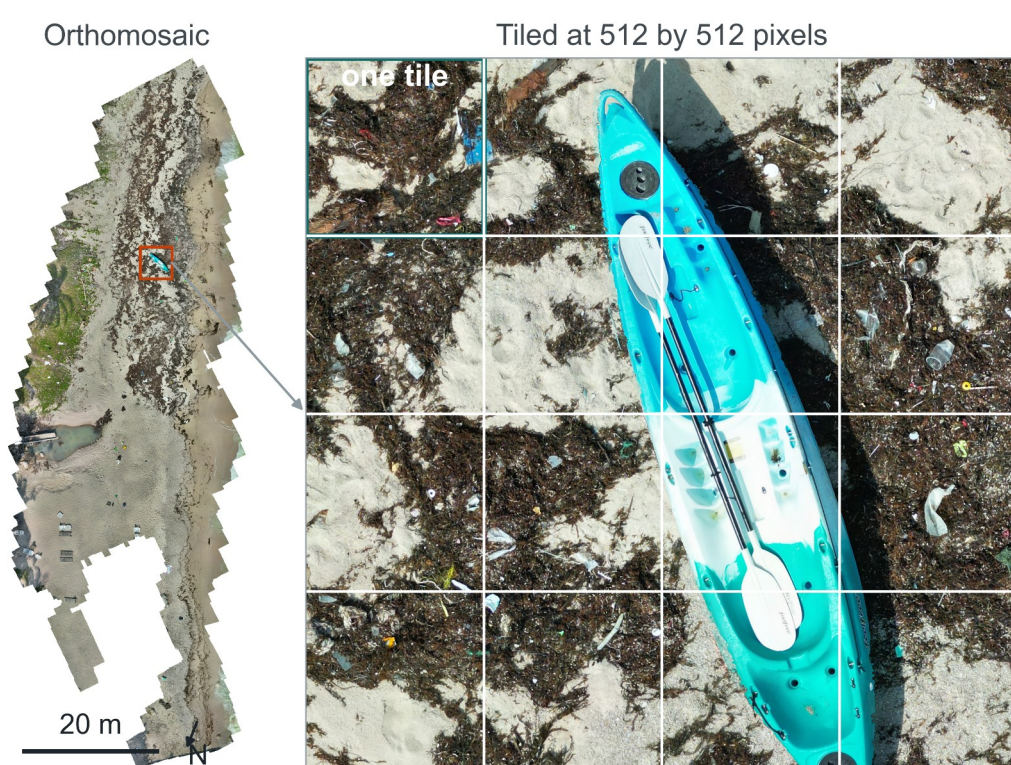


Fig. 3. Orthomosaic and 512-pixel tiling at native resolution.

PLAS-Net Architecture

Pixel-level Litter Area Segmentor: three feature-fusion and attention modules on a YOLO26n-seg baseline.

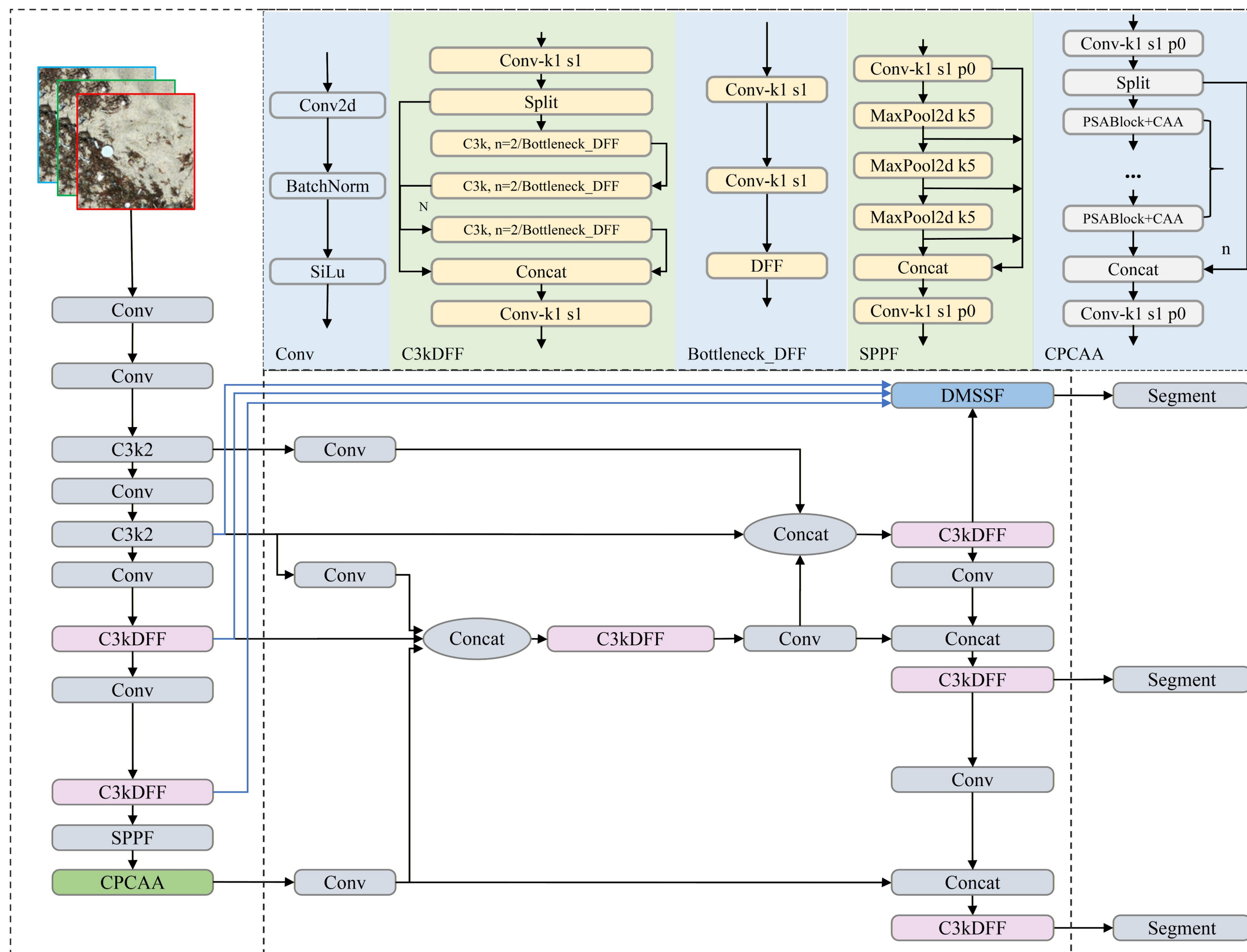


Fig. 4. PLAS-Net, with C3kDFF, CPCA and DMSSF shown in place.

C3kDFF

Sand, shells and wrack mimic weathered debris.

Channel and spatial gates damp those responses before fusion.

CPCA

Line and netting run across scales and are often occluded.
Strip convolutions link separated parts of one object.

DMSSF

Items range from small fragments to large domestic objects.
Dynamic resampling keeps small and elongated items through the merge.

Our Contributions

- UAV orthomosaic at 0.17 cm per pixel, polygon-annotated across 13 categories
- Three modules on YOLO26n-seg, 1.5 % parameter cost (2.69 to 2.73 M)
- Benchmarked against 11 segmentation models, plus a cumulative ablation
- Mask-derived projected area: count and footprint rankings differ

Discussion and Limitations

- 145 FPS: forward pass only, batch 1, FP32, RTX 5090
- Offline screening of tiled orthomosaics, not onboard real-time
- Recall 46.7 to 44.4: precision favoured, buried items still missed
- Projected area: application demonstration, not a validated measurement
- One beach, one survey: generalization untested

Comparison with Segmentation Models

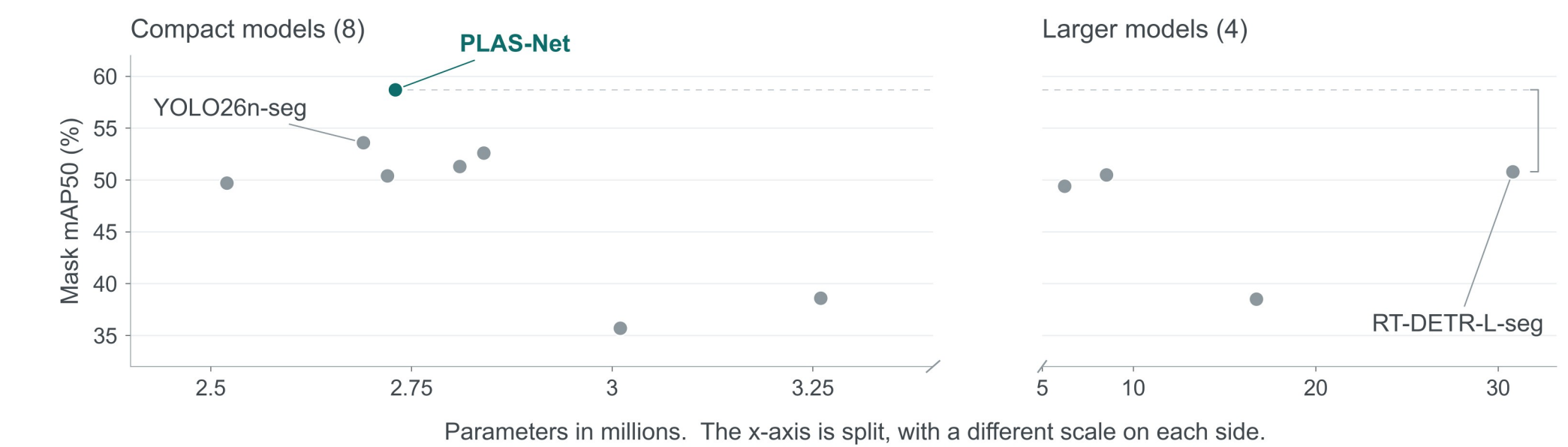


Fig. 5. Mask mAP50 against parameter count, all twelve models.

Model	Params	mAP50	mAP50:95
RT-DETR-L-seg	30.81	50.8	41.8
YOLOv8n-seg	3.26	38.6	30.3
YOLOv11n-seg	2.84	52.6	43.9
YOLO26n-seg	2.69	53.6	44.3
PLAS-Net	2.73	58.7	46.8

Table 2. Selected models on the 395-image evaluation split.

Cumulative Ablation

	YOLO26n-seg	+ C3kDFF	+ CPCA	+ DMSSF (PLAS-Net)
Mask mAP50 (%)	53.6	55.5 (+1.9 pp)	57.8 (+2.3 pp)	58.7 (+0.9 pp)
Mask precision (%)	53.2	63.5	65.4	69.7
Mask recall (%)	46.7	42.4	46.8	44.4

Fig. 6. Cumulative ablation of the three modules.

Qualitative Results

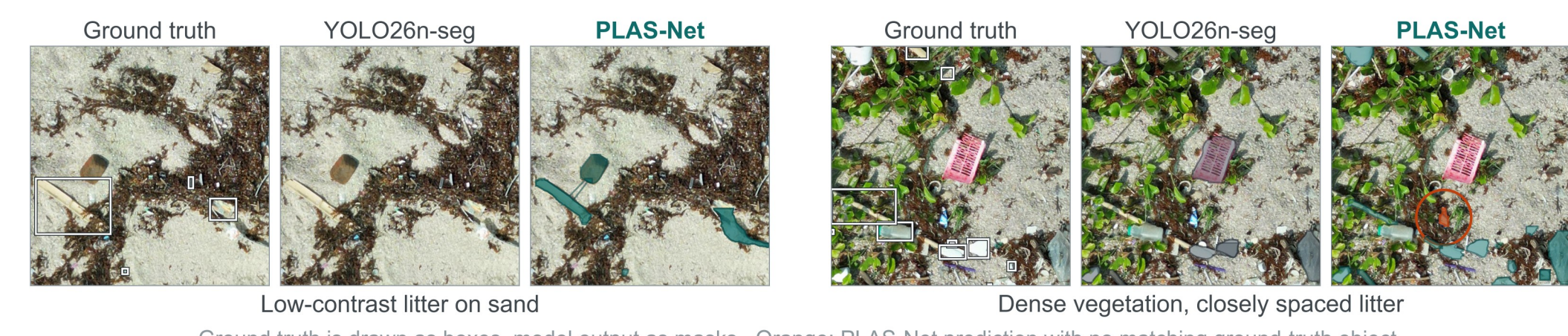


Fig. 7. Two difficult scenes; one unmatched prediction circled.

Mask-Derived Summaries

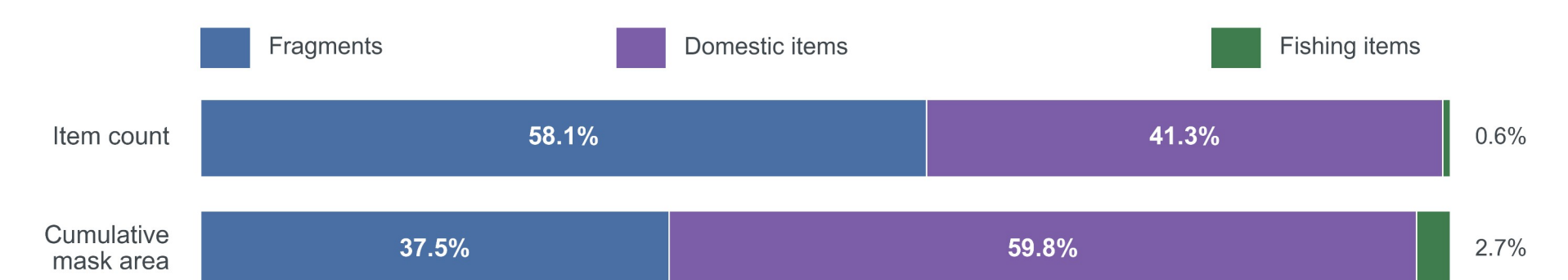


Fig. 8. Count and mask-area shares by source group; 2,957 instances scene-wide.